LUNG VOLUMES AND CAPACITIES

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Learning objectives

At the end of lecture the students should be able to;

1. **Define & describe different lung volumes including**
   - Tidal Volume (TV)
   - Inspiratory Reserve Volume (IRV)
   - Expiratory Reserve Volume (ERV)
   - Residual Volume (RV).

2. **Illustrate different lung capacities.**
3. **Describe the dead space & its types.**
4. **Explain PFTs.**
5. **Describe the significance of PFTs.**

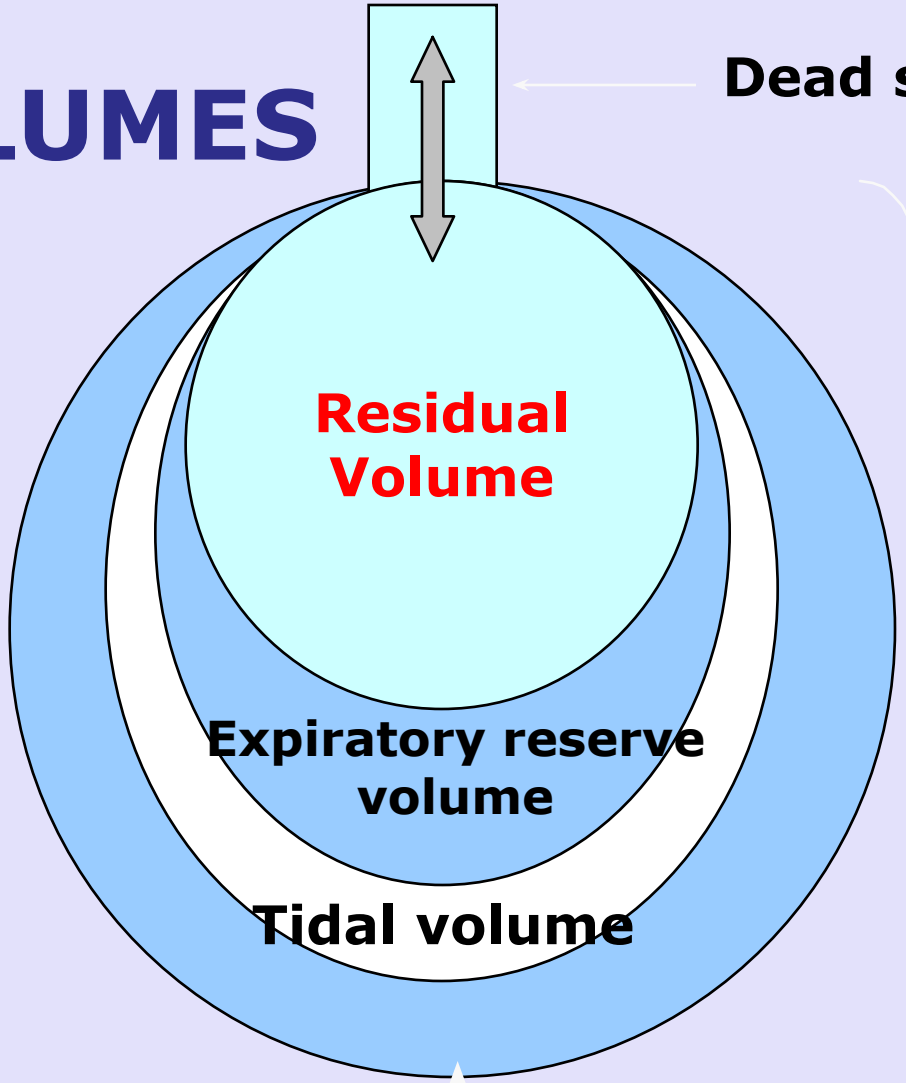
Volumes Versus Capacities

- There are **four volume subdivisions** which:
  - do not overlap.
  - can not be further divided.
  - when added together equal total lung capacity.
- **Lung capacities** are subdivisions of total volume that include two or more of the 4 basic lung volumes.

Basic lung volumes

- Tidal Volume (TV)
- Inspiratory Reserve Volume (IRV)
- Expiratory Reserve Volume (ERV)
- Residual Volume (RV).

LUNG VOLUMES



Dead Space

- Dead space is the volume of the airways and lungs that does not participate in gas exchange.

TYPES OF DEAD SPACE

- Dead space is of two types:
  1. Anatomical dead space
  2. Physiological dead space
- Volume of normal dead space is 150 mL. Under normal conditions, physiological dead space is equal to anatomical dead space. It is because, all the alveoli are functioning and all the alveoli receive adequate blood flow in normal conditions.

- **Anatomical Dead Space** extends from nose up to terminal bronchiole. It includes nose, pharynx, trachea, bronchi and branches of bronchi up to terminal bronchioles. These structures serve only as the passage for air movement. Gaseous exchange does not take place in these structures.

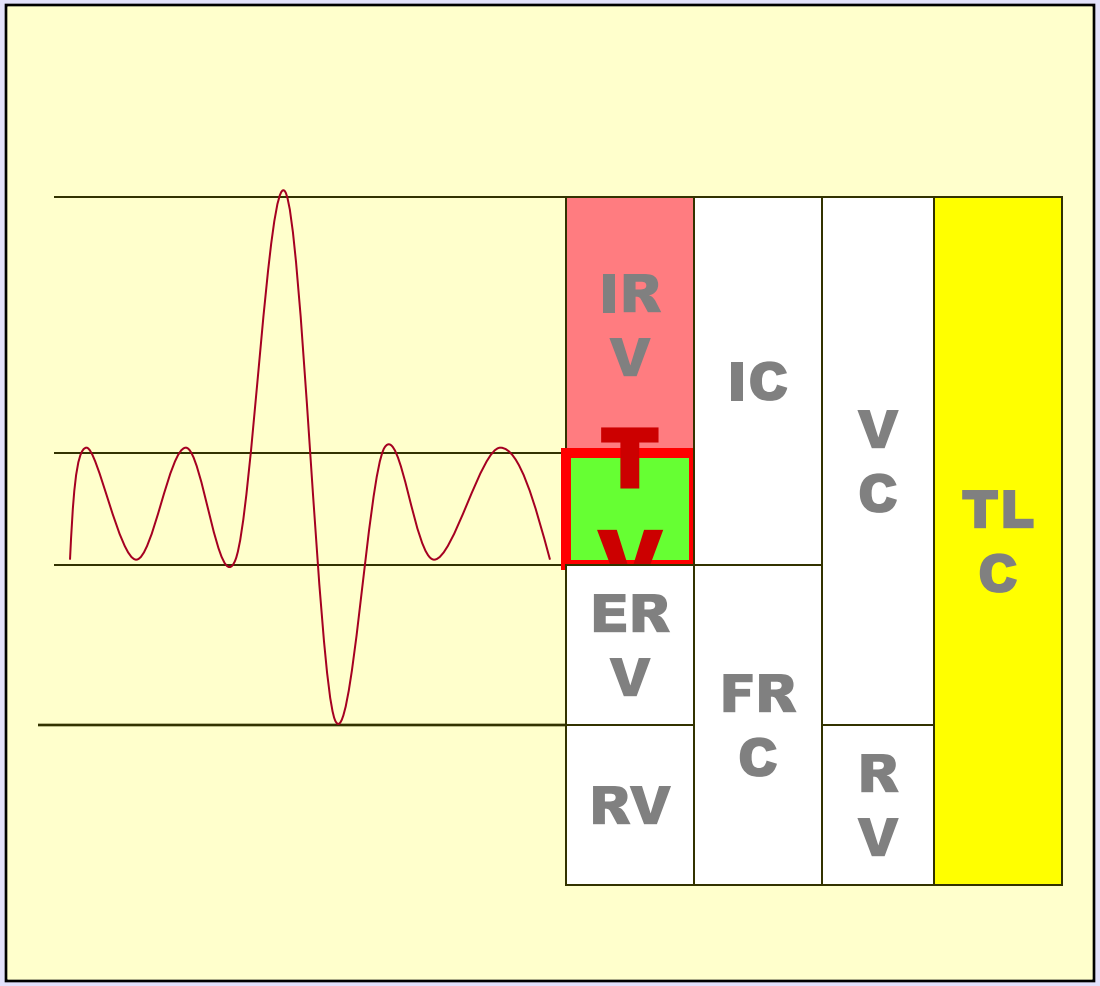
Physiological Dead Space

- **Physiological Dead Space** includes anatomical dead space plus two additional volumes. Additional volumes are :

  1. Air in the alveoli, which are non-functioning. In some respiratory diseases, alveoli do not function because of dysfunction or destruction of alveolar membrane.

  2. Air in the alveoli, which do not receive adequate blood flow. Gaseous exchange does not take place during inadequate blood supply. These two additional volumes are generally considered as wasted ventilation.

Tidal Volume (TV)

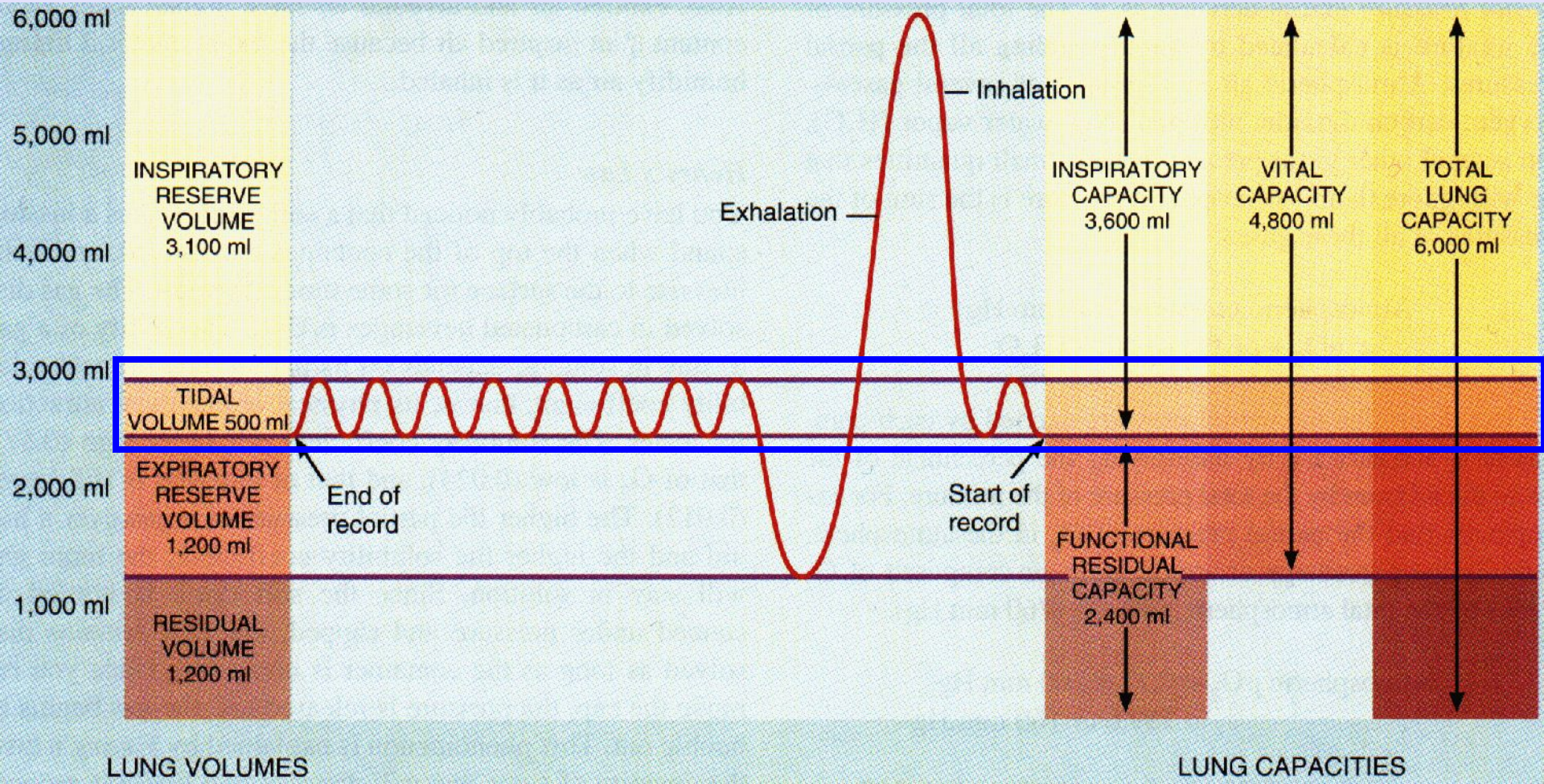


- The volume of gas inspired or expired with each breath.

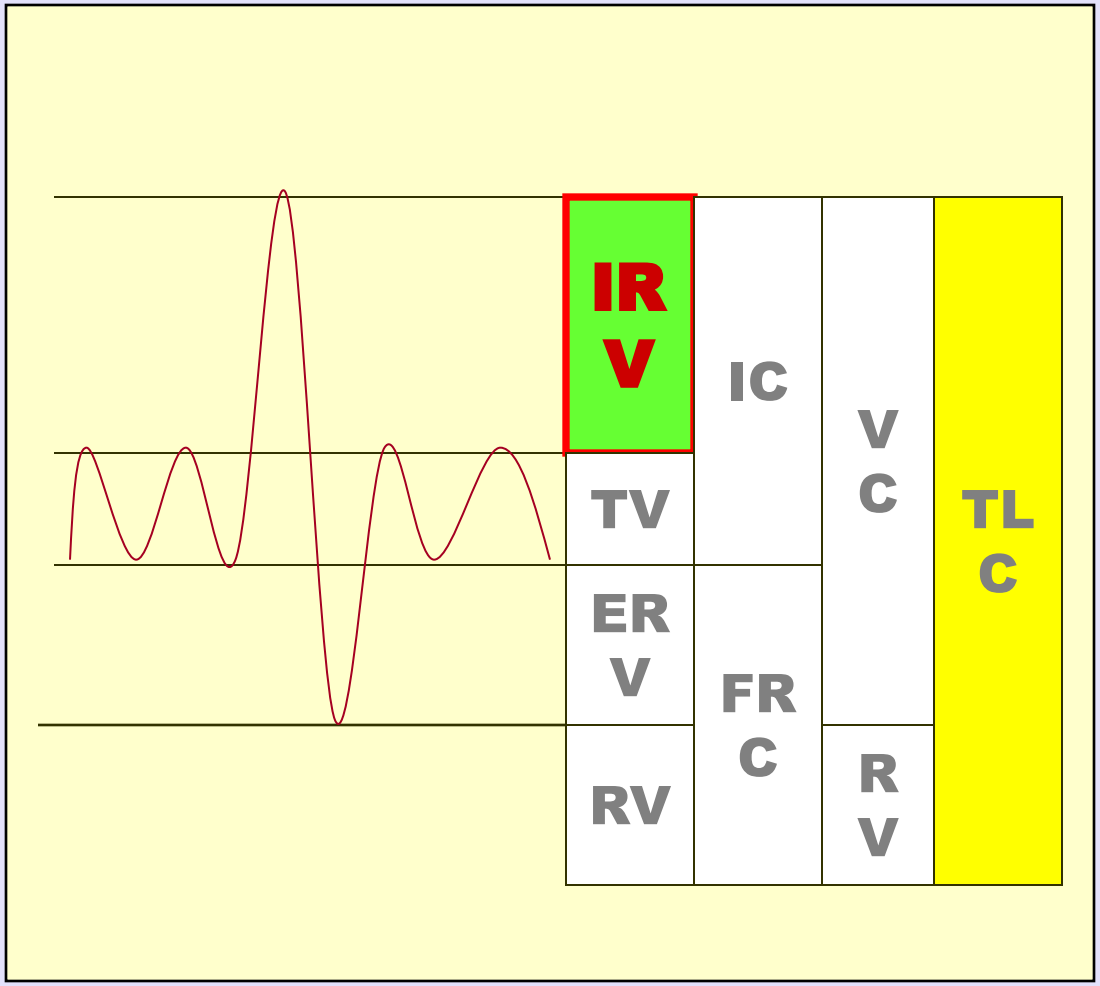
  during normal quiet breathing

Tidal Volume (V_T)

- volume of one breath
- about 500 ml
- only about 70% (350ml) actually reaches alveoli



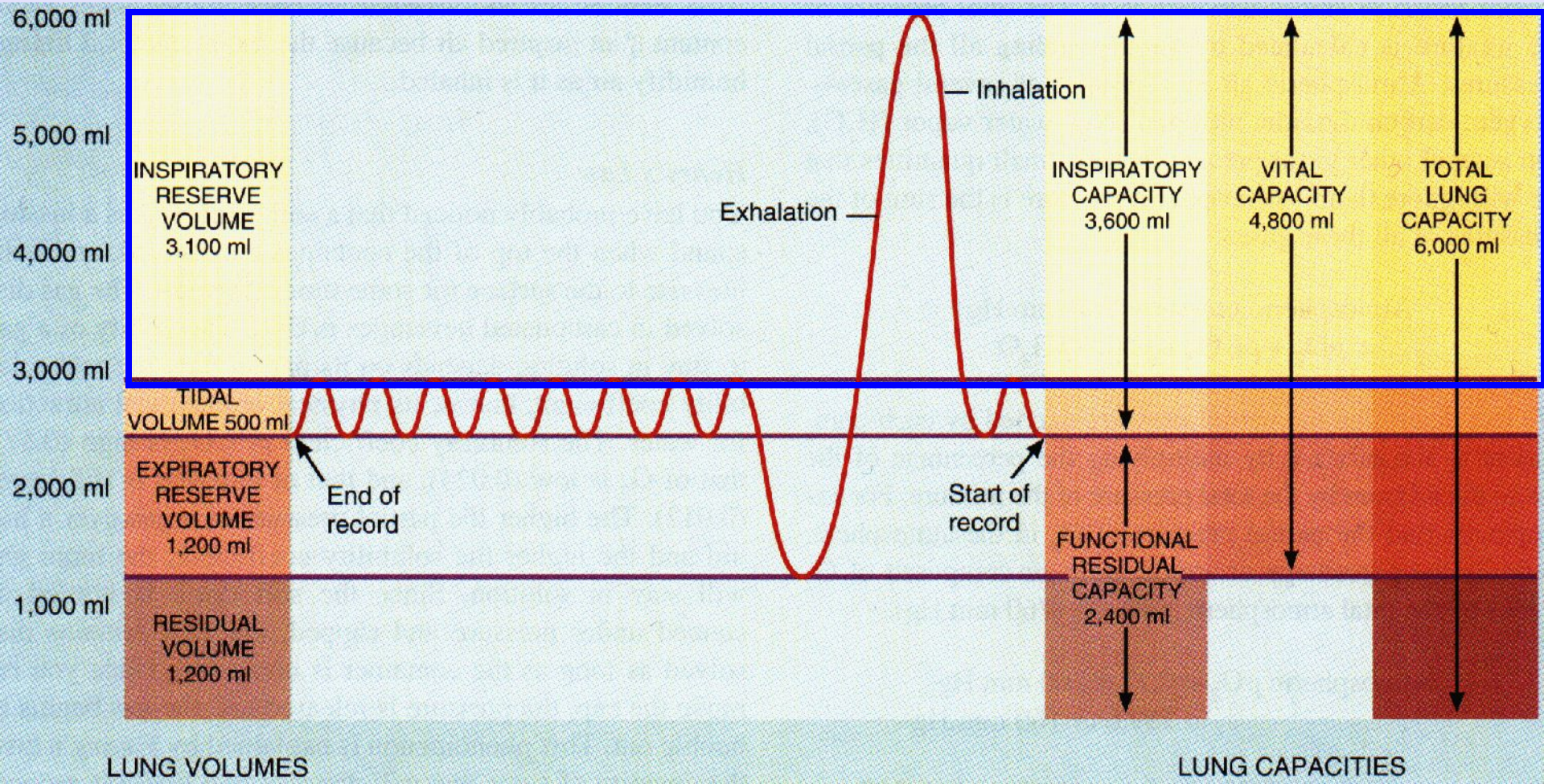
Inspiratory Reserve Volume (IRV)



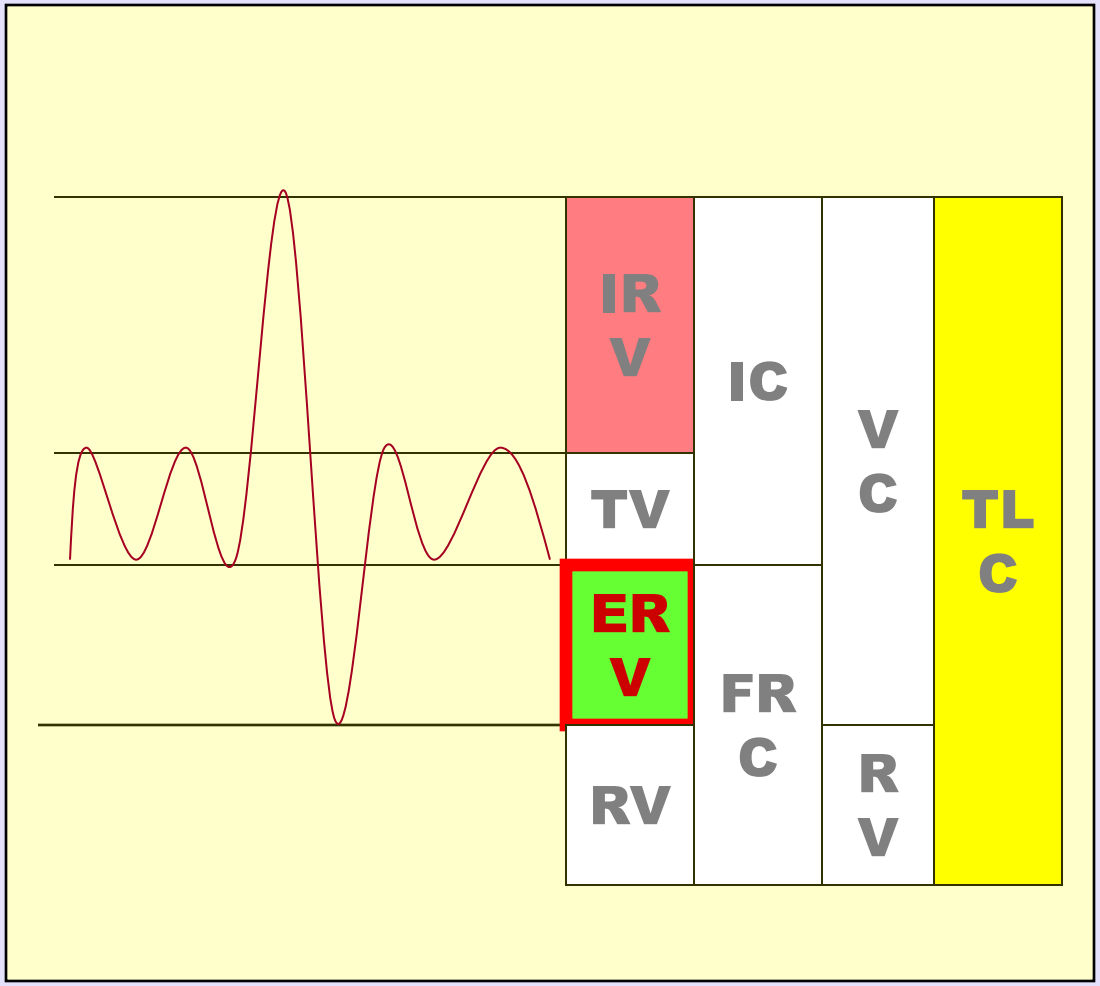
- The maximum amount of air that can be inhaled after a normal tidal volume inspiration

Inspiratory Reserve Volume (IRV)

- air inspired above normal quiet breathing
- 3100ml above 500ml V_T



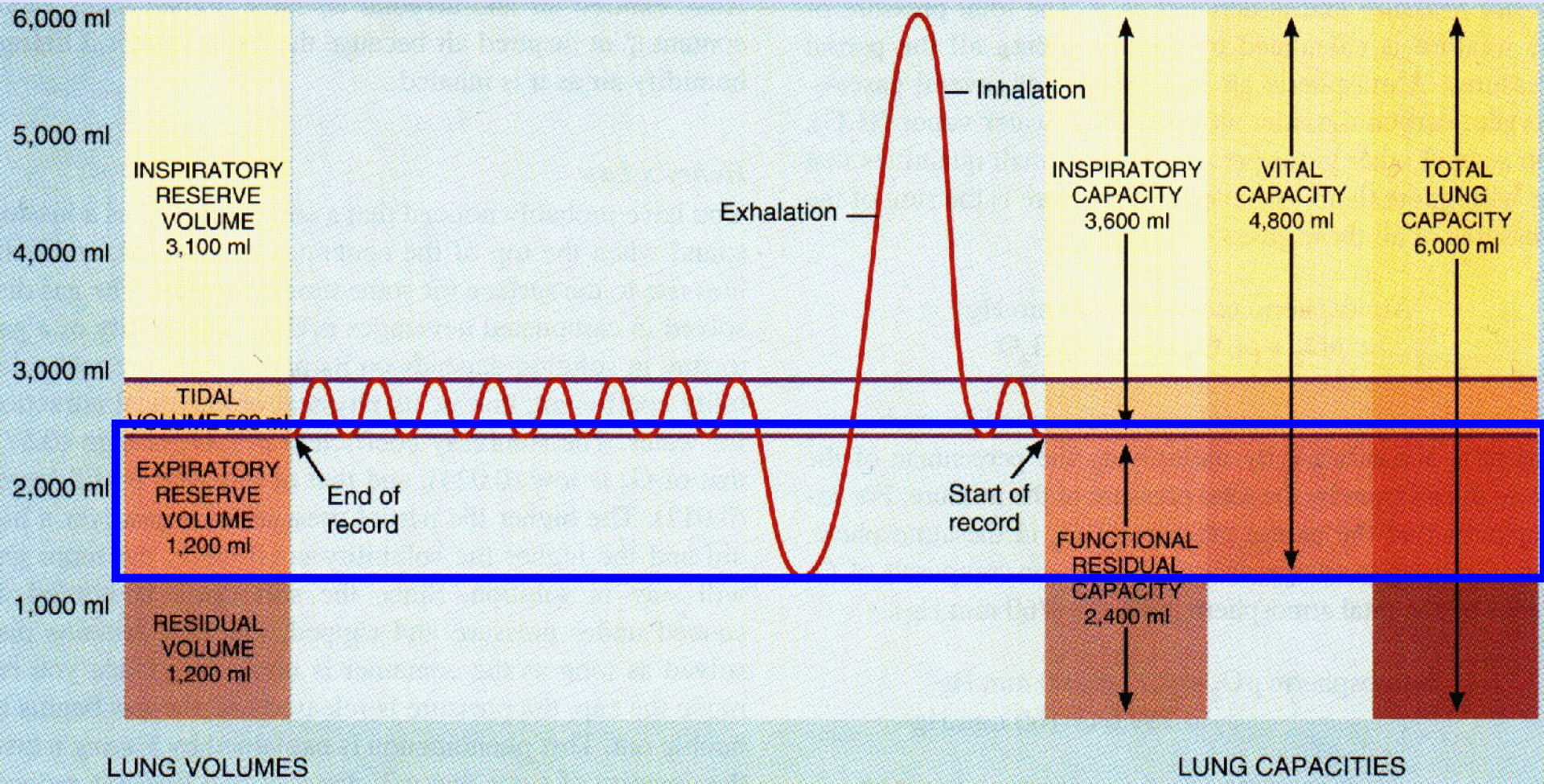
Expiratory Reserve Volume (ERV)



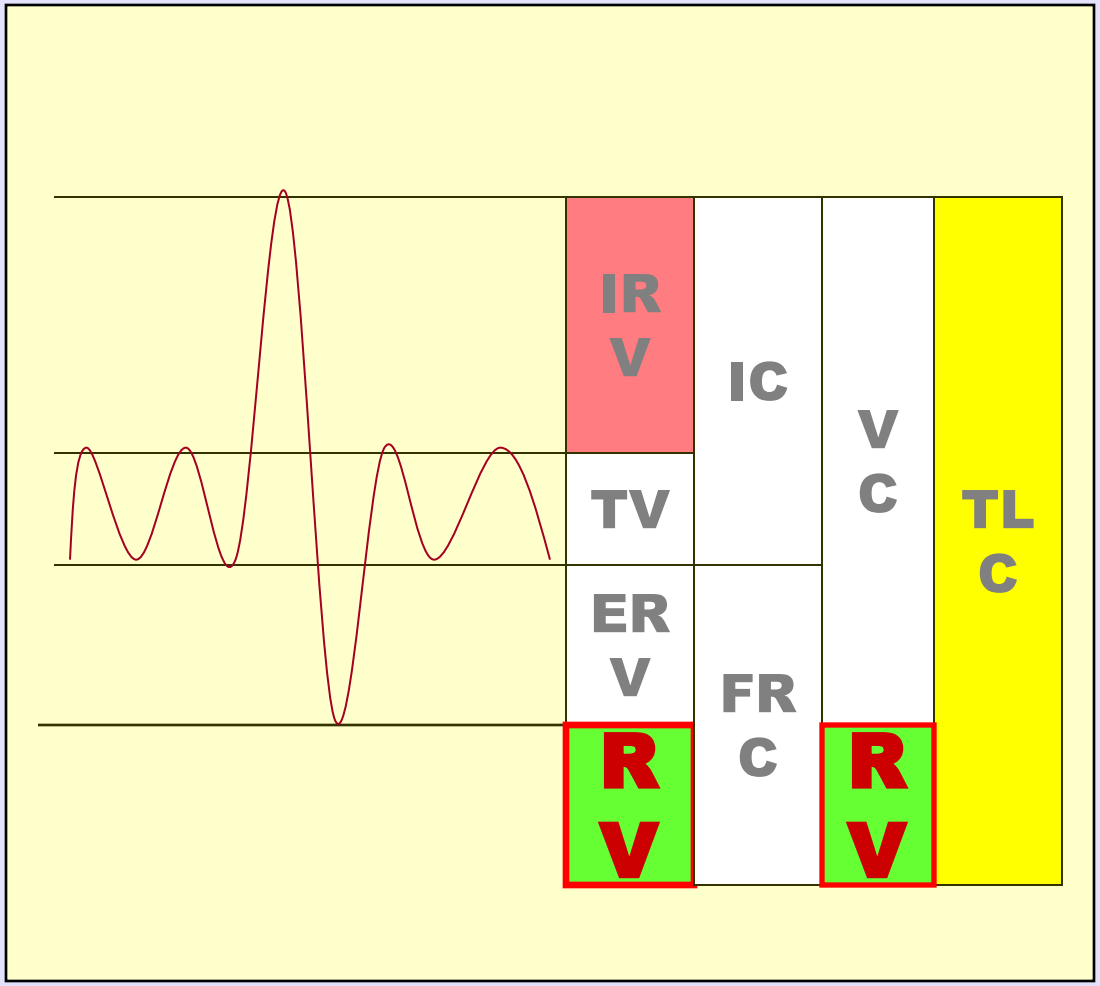
- Maximum amount of air that can be exhaled from the resting expiratory level

Expiratory Reserve Volume (ERV)

- air expired above normal quiet breathing
- 1200 ml above 500ml V_T



Residual Volume (RV)



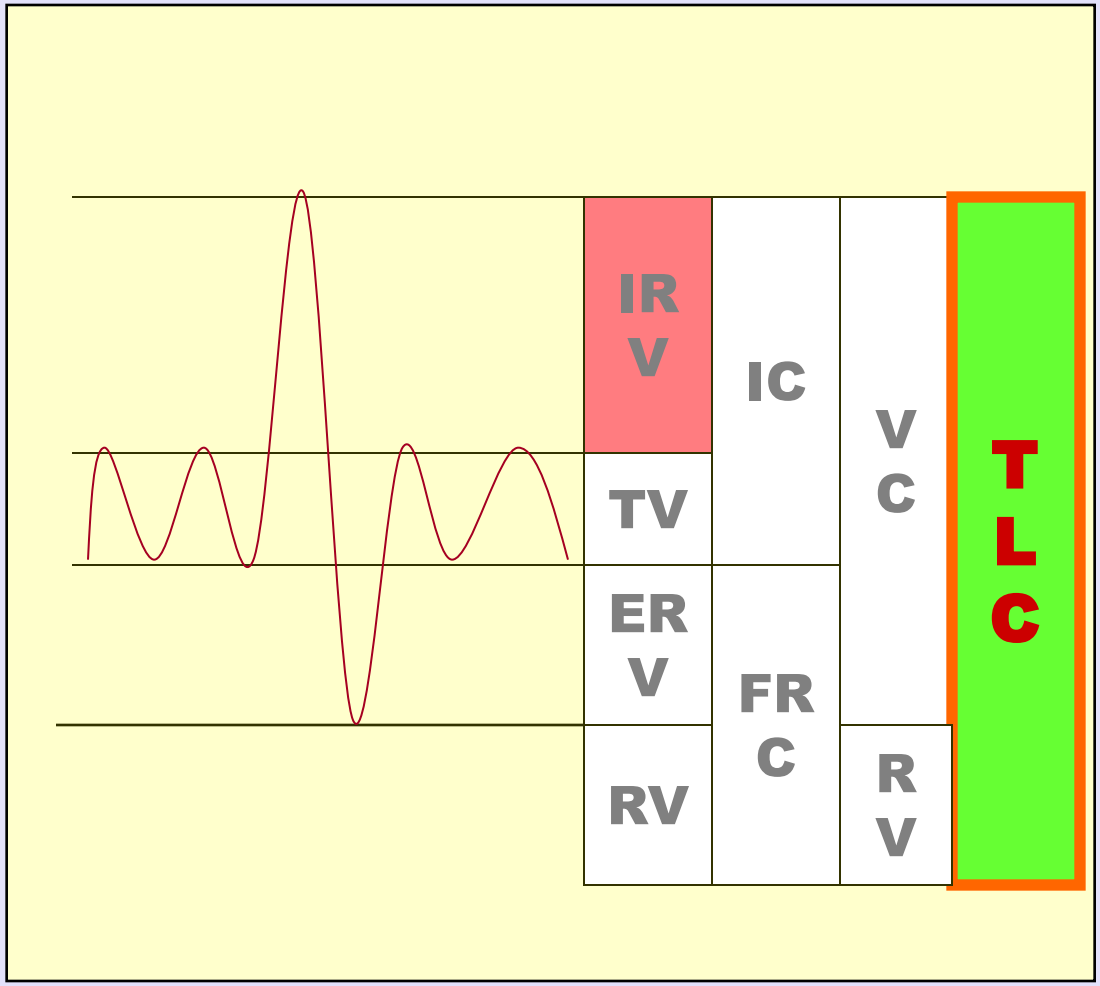
The volume of air remaining in the lung after a maximal expiration.

This is the only lung volume which cannot be measured with a spirometer.

Basic lung capacities

- Total Lung Capacity (TLC).
- Vital Capacity (VC).
- Functional Residual Capacity (FRC).
- Inspiratory Capacity (IC).

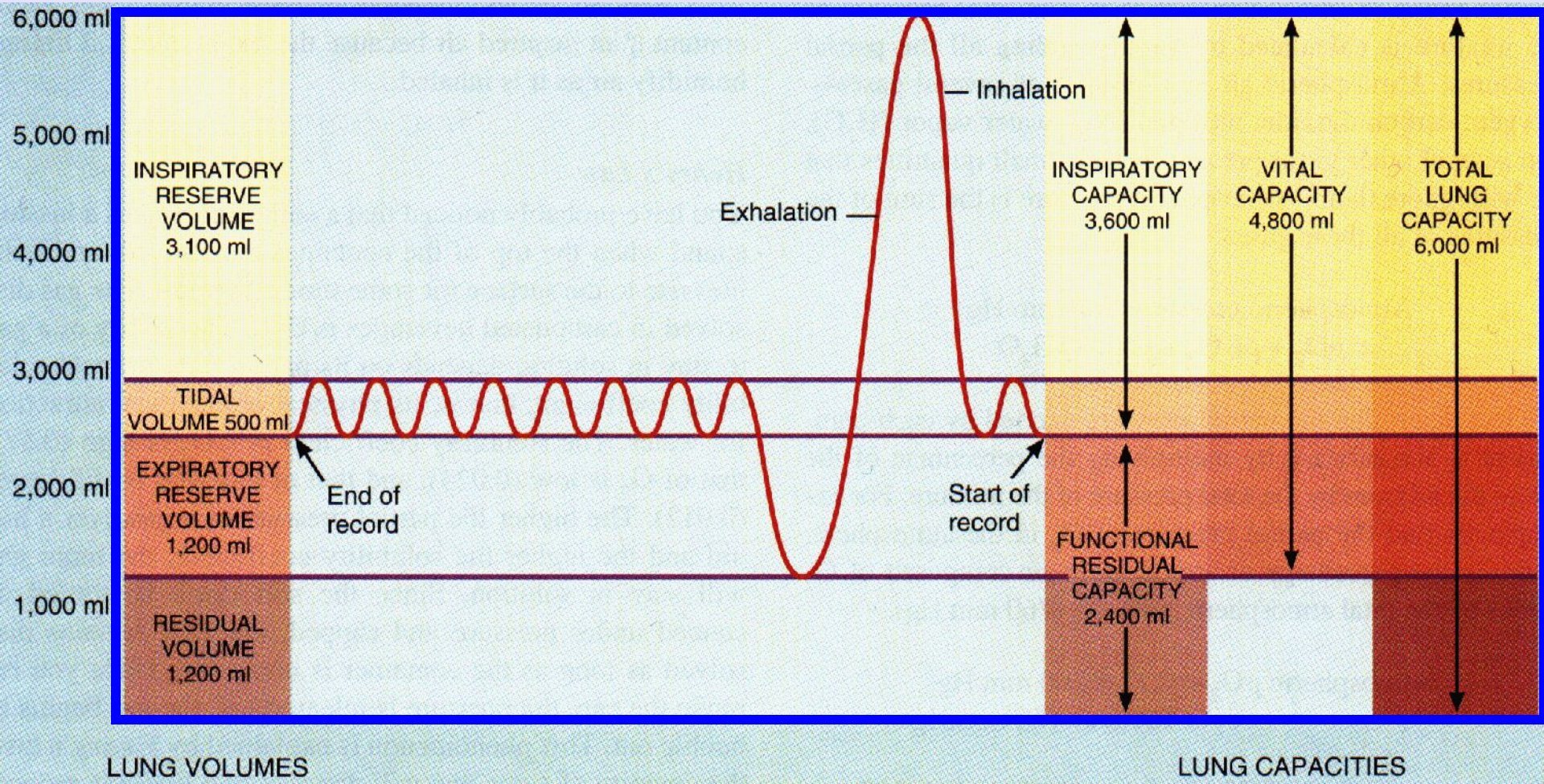
Total Lung Capacity (TLC)



- The volume of air contained in the lungs at the end of a maximal inspiration.
- Sum of all four basic lung volumes
- TLC = RV + IRV + TV + ERV

Total Lung Capacity (TLC)

- **TLC= IRV + V_T + ERV + RV = 6000ml**



MEASUREMENT OF LUNG VOLUMES AND CAPACITIES

- Spirometry is the method to measure lung volumes and capacities. Simple instrument used for this purpose is called spirometer.
- Nowadays plethysmograph is also used to measure lung volumes and capacities.
- Residual volume cannot be measured by spirometry, lung capacities that include the residual volume also cannot be measured by spirometry (i.e., FRC and TLC). Two methods are used to measure FRC: helium dilution and the body plethysmograph.

Total Lung Capacity (TLC)

- Measured by
  - **body plethysmography**
  - **helium dilution**
- **Body plethysmography**
  - mouthpiece obstructed with shutter
  - rapid panting
  - chest volume expand and decompress the air in the lungs
  - changes in pressure inside the box allow determination of the lung volume



By applying Boyle's law ($P \cdot V$ = constant) ☐ TLC

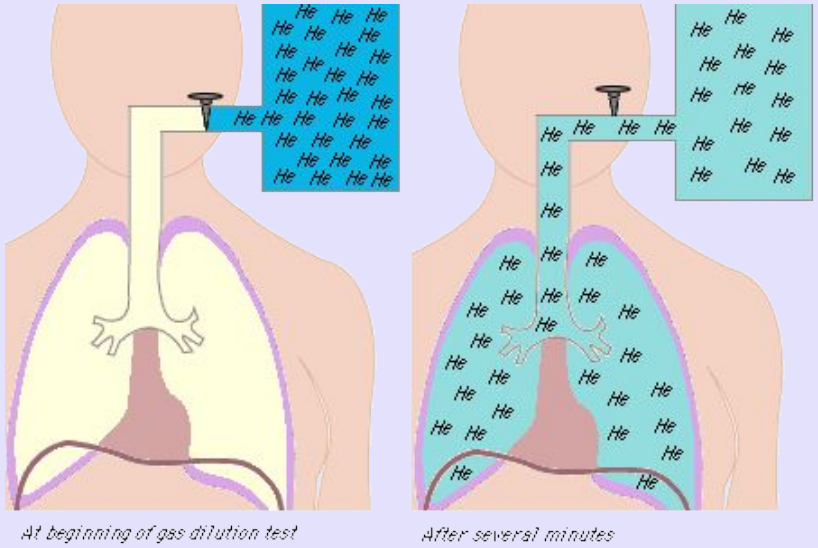
Total Lung Capacity (TLC)

- **Helium dilution**

- Spirometer of known volume and helium concentration connected to the patient

- Closed circuit

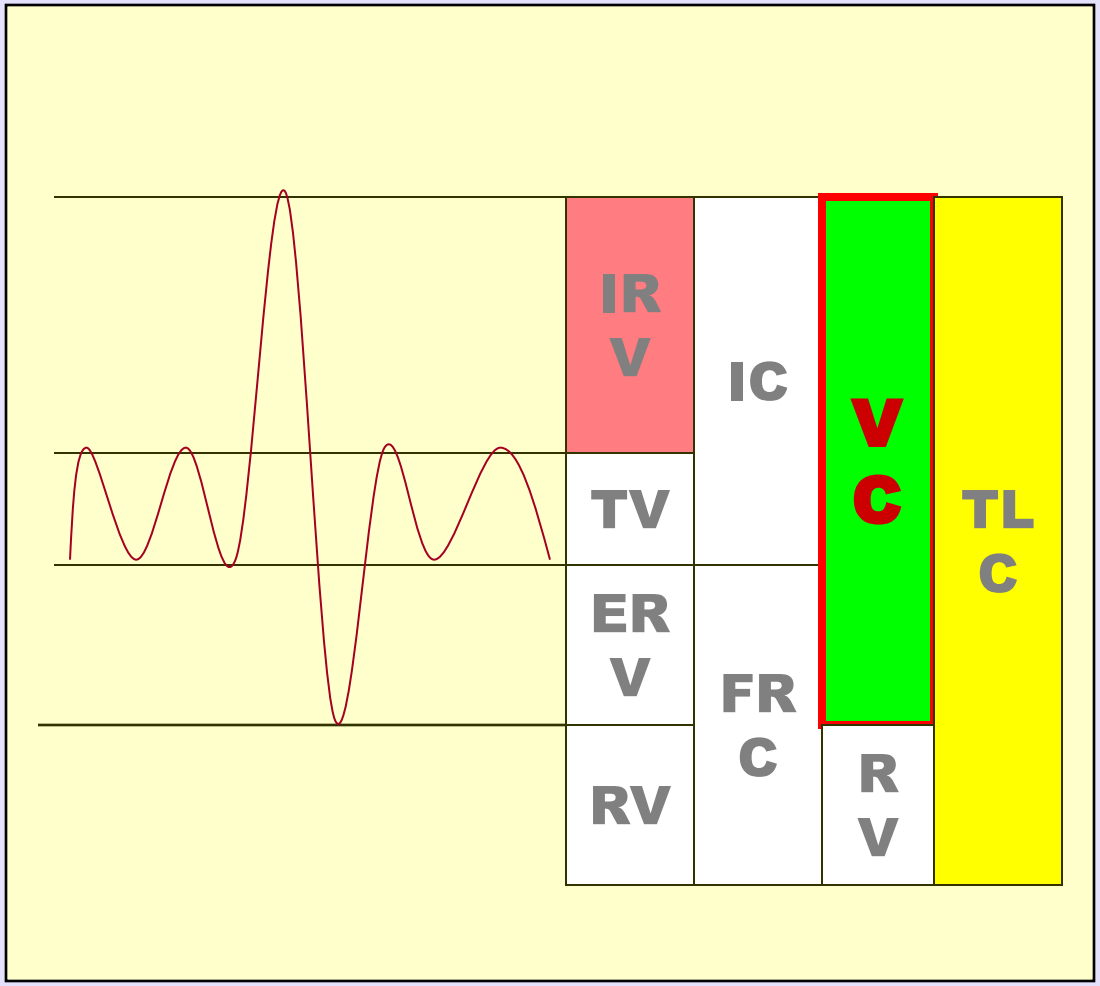
- Law of conservation of mass



$$[He]_{initial} \cdot V_s = [He]_{final} \cdot (V_s + V_L)$$

Unknown lung volume can be calculated

Vital Capacity (VC)

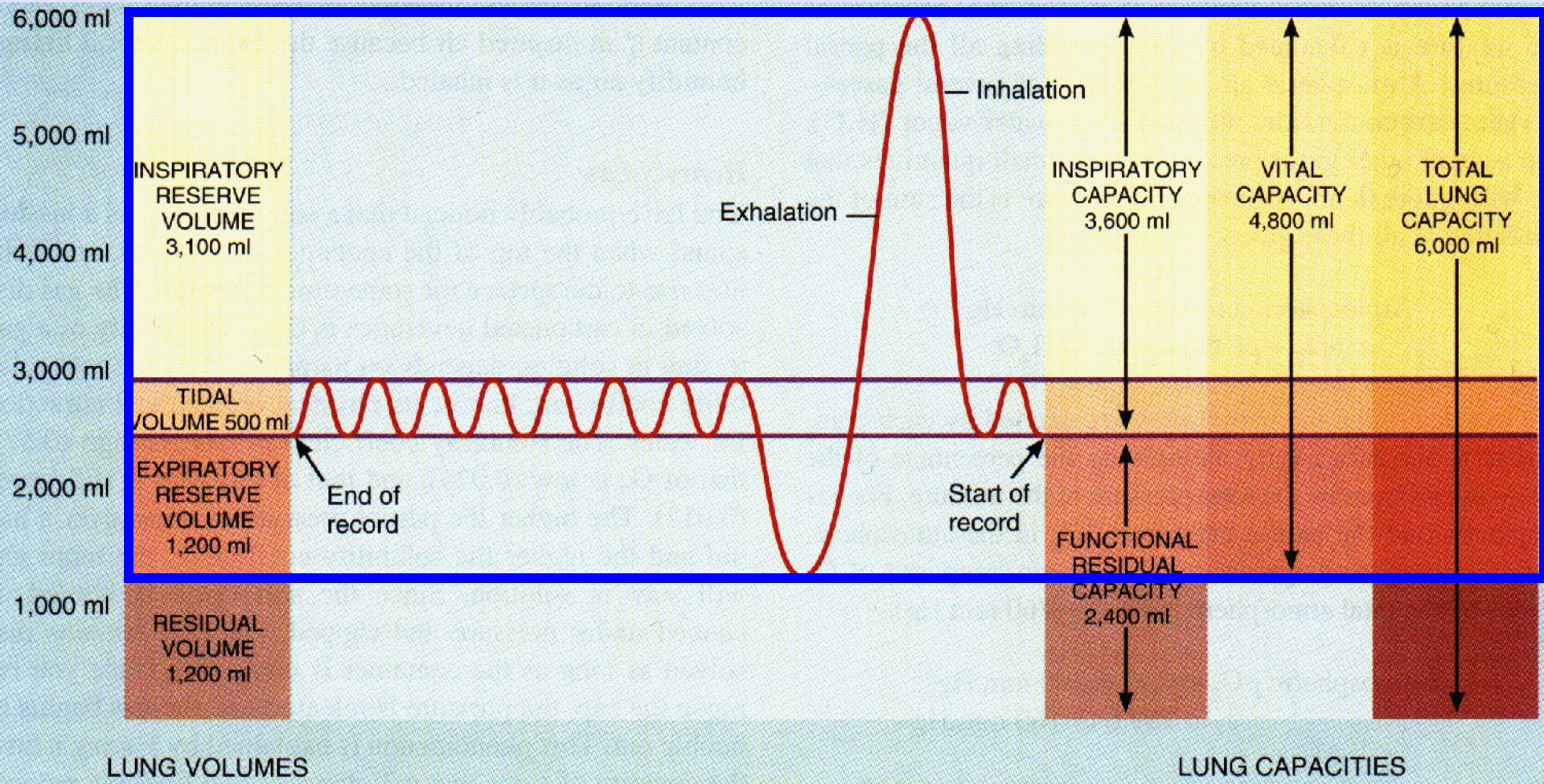


- The maximum volume of air that can be forcefully expelled from the lungs following a maximal inspiration.
- Largest volume that can be measured with a spirometer
- **VC= IRV + TV + ERV**

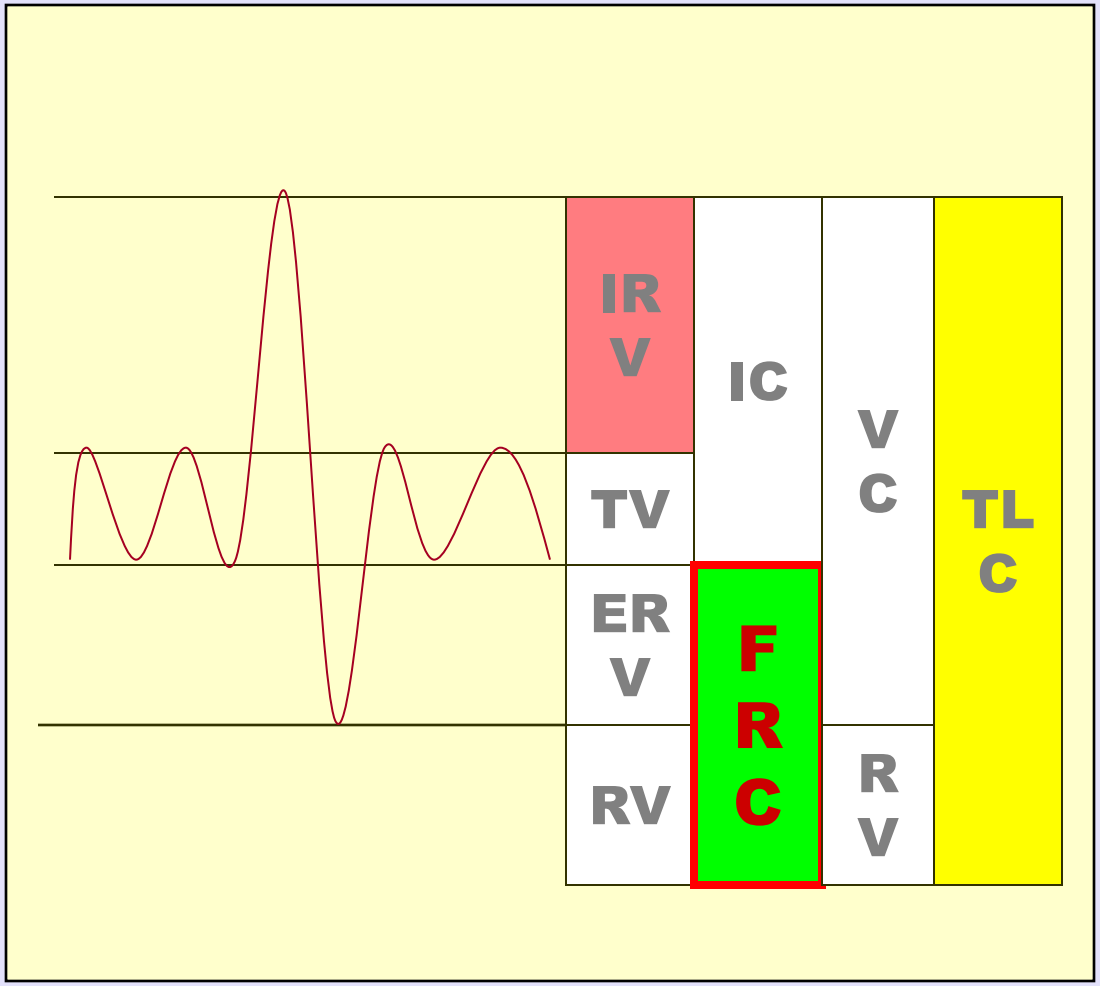
  = TLC - RV

Vital Capacity (VC)

- **VC = IRV + V_T + ERV = 4800 ml**

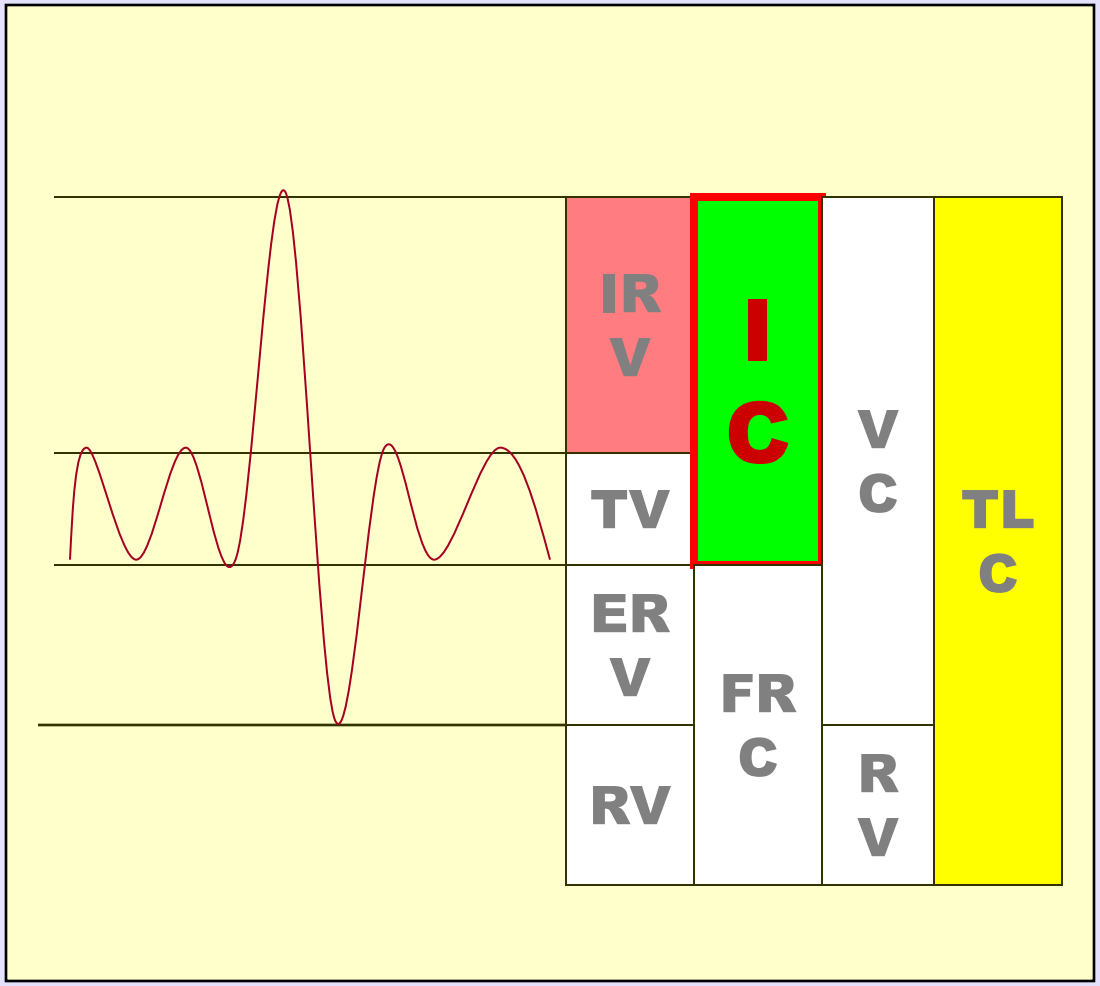


Functional Residual Capacity (FRC)



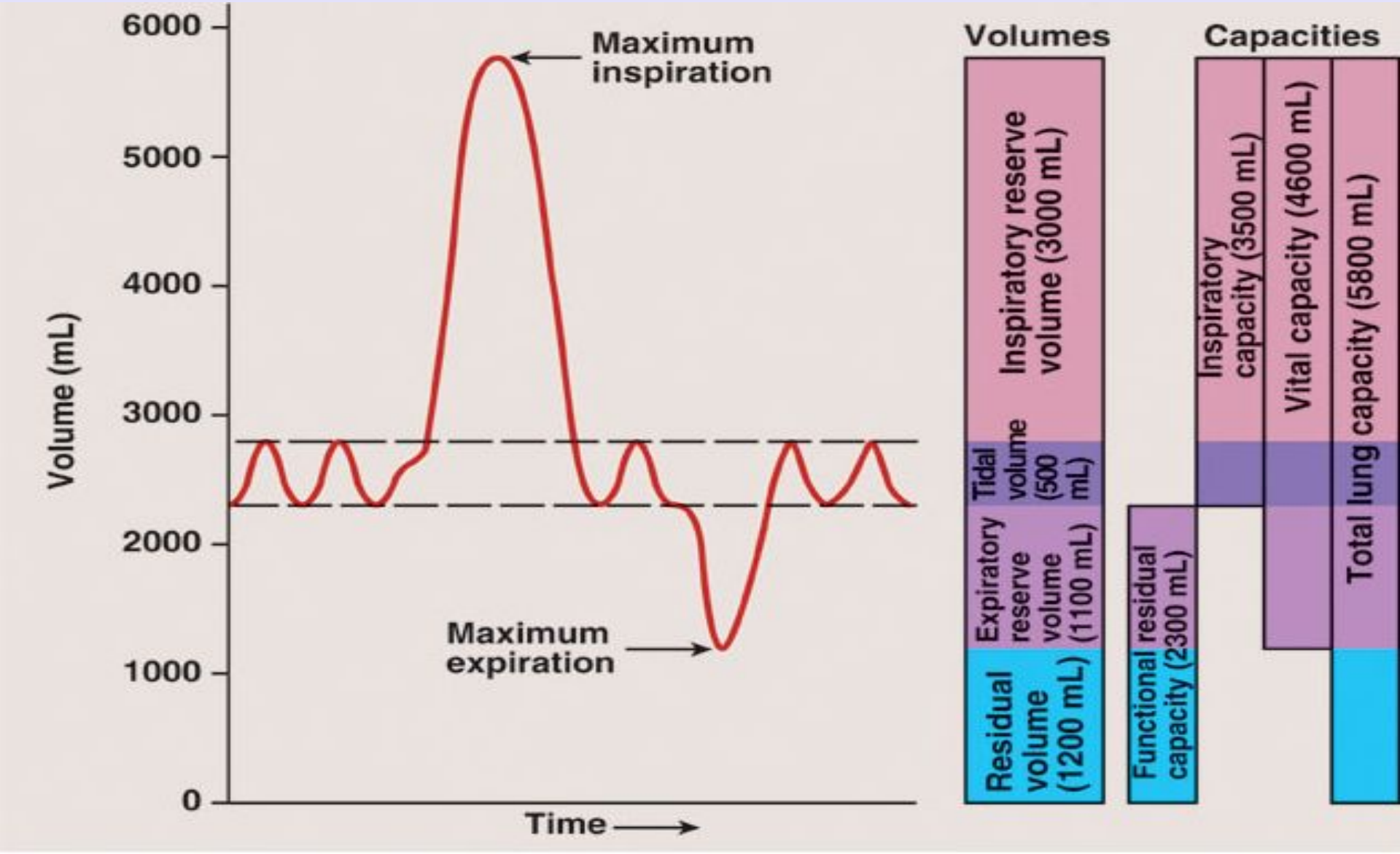
- Volume of air remaining in the lungs at the end of a normal TV expiration
- The elastic force of the chest wall is exactly balanced by the elastic force of the lungs
- **FRC = ERV + RV**

Inspiratory Capacity (IC)



- Maximum volume of air that can be inspired from end expiratory position.
- This capacity is of less clinical significance than the other three.
- **IC = TV + IRV**

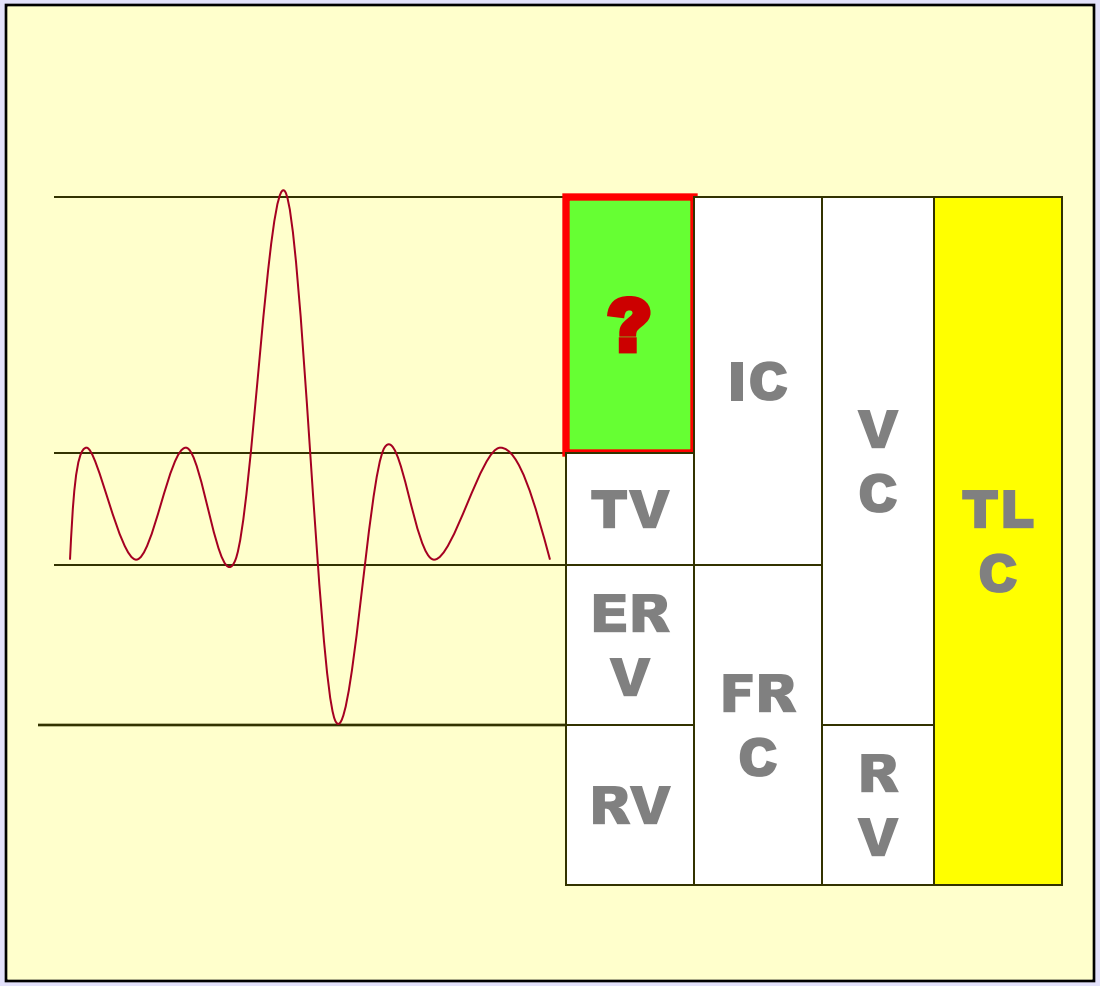
LUNG VOLUMES AND CAPACITIES



SELF ASSESSMENT

Q1. FRC = ERV + ------?

Q2. Name the volume marked in the diagram and give its value.



Pulmonary Function Tests (PFTs)

- Pulmonary function tests or lung function tests are useful in assessing the functional status of the respiratory system both in physiological and pathological conditions.
- Lung function tests are based on the measurement of volume of air breathed in and out in quiet breathing and forced breathing. These tests are carried out mostly by using spirometer.

TYPES OF LUNG FUNCTION TESTS

Lung function tests are of two types:

1. Static lung function tests
2. Dynamic lung function tests.

Static Lung Function Tests

- Static lung function tests are based on volume of air that flows into or out of lungs. These tests do not depend upon the rate at which air flows. Static lung function tests include static lung volumes and static lung capacities.

Dynamic Lung Function Test

- Dynamic lung function tests are based on time, i.e. the rate at which air flows into or out of lungs. These tests include forced vital capacity, forced expiratory volume, maximum ventilation volume and peak expiratory flow. Dynamic lung function tests are useful in determining the severity of obstructive and restrictive lung diseases.

RESTRICTIVE AND OBSTRUCTIVE RESPIRATORY DISEASES

Diseases of respiratory tract are classified into two types:

1. Restrictive respiratory disease
2. Obstructive respiratory disease.

These two types of respiratory diseases are determined by lung functions tests, particularly FEV.

RESTRICTIVE RESPIRATORY DISEASE

Restrictive respiratory disease is the abnormal respiratory condition characterized by difficulty in inspiration. Expiration is not affected. Restrictive respiratory disease may be because of abnormality of lungs,thoracic cavity or/and nervous system.

OBSTRUCTIVE RESPIRATORY DISEASE

Obstructive respiratory disease is the abnormal respiratory condition characterized by difficulty in expiration.

TABLE 121.1: Restrictive and obstructive respiratory diseases

Type	Disease	Structures involved
Restrictive respiratory diseases	Polio myelitis	CNS
	Myasthenia gravis	CNS and thoracic cavity
	Flail chest (broken ribs)	Thoracic cavity
	Paralysis of diaphragm	CNS
	Spinal cord diseases	CNS
	Pleural effusion	Thoracic cavity
Obstructive respiratory diseases	Asthma Chronic bronchitis Emphysema Cystic fibrosis	Lower respiratory tract
	Laryngotracheobronchitis Epiglottis Tumors Severe cough and cold with phlegm	Upper respiratory tract

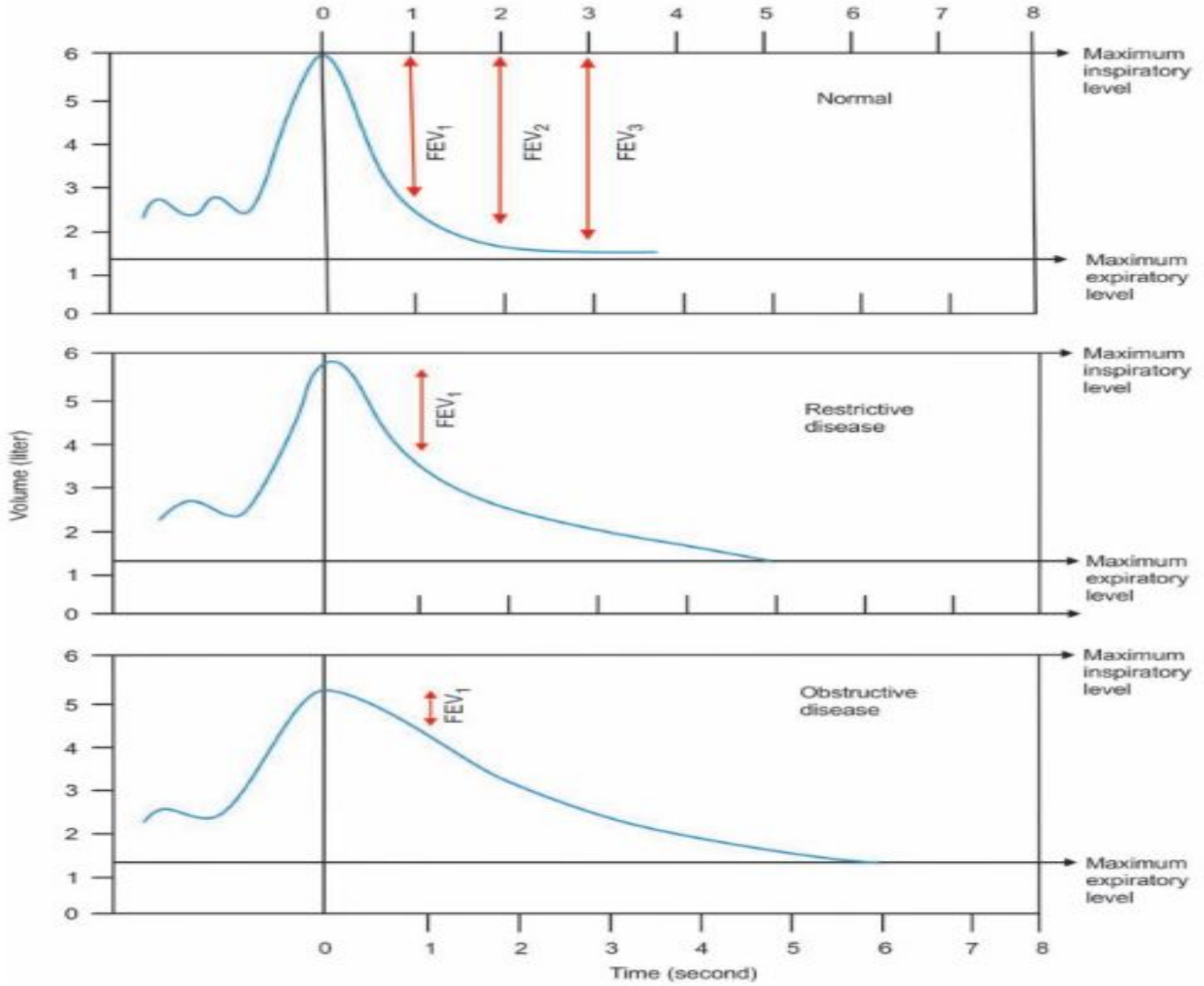


FIGURE 121.5: Forced expiratory volume. FEV = Forced expiratory volume.

FORCED VITAL CAPACITY

Forced vital capacity (FVC) is the volume of air that can be exhaled forcefully and rapidly after a maximal or deep inspiration. It is a dynamic lung capacity.

Normally FVC is equal to VC. However in some pulmonary diseases, FVC is decreased.

FORCED EXPIRATORY VOLUME OR TIMED VITAL CAPACITY

Forced expiratory volume (FEV) is the volume of air,which can be expired forcefully in a given unit of time (after a deep inspiration). It is also called timed vital capacity or forced expiratory vital capacity (FEVC). Itis a dynamic lung volume.

- FEV1= Volume of air expired forcefully in 1 second
- FEV2= Volume of air expired forcefully in 2 seconds
- FEV3= Volume of air expired forcefully in 3 seconds.

NORMAL VALUES

Forced expiratory volume in persons with normal respiratory functions is as follows:

- FEV1= 83% of total vital capacity
- FEV2= 94% of total vital capacity
- FEV3= 97% of total vital capacity
- After 3rd second = 100% of total vital capacity.

SIGNIFICANCE OF DETERMINING FEV

- Vital capacity may be almost normal in some of the respiratory diseases. However, the FEV has great diagnostic value, as it is decreased significantly in some respiratory diseases.
- It is very much decreased in obstructive diseases like asthma and emphysema.
- It is slightly reduced in some restrictive respiratory diseases like fibrosis of lungs.


Thank You